BOOK RECOMMENDATIONS SYSTEMS PROJECT

collaboration Filtering based Recommendations system

```
import pandas as pd
import numpy as np
```

```
books = pd.read_csv('Books.csv')
ratings = pd.read_csv('Ratings.csv')
users = pd.read_csv('Users.csv')
```

```
books.head()
```

Show hidden output

```
users.head()
```

Show hidden output

```
ratings.head()
```

Show hidden output

Shape of Dataset ❓

```
print("The shape of book is    : ",books.shape)
print("The shape of ratings is : ",ratings.shape)
print("The shape of users is   : ",users.shape)
```

```
The shape of book is    :  (101807, 8)
The shape of ratings is :  (1149780, 3)
The shape of users is   :  (278858, 3)
```

Nan value in Dataset ❓

```python
books.isnull().sum()
```

Show hidden output

```python
users.isnull().sum()
```

	0
User-ID	0
Location	0
Age	110762

dtype: int64

```python
ratings.isnull().sum()
```

	0
User-ID	0
ISBN	0
Book-Rating	0

dtype: int64

Duplication in Dataset ❓

```python
books.duplicated().sum()
```

```
np.int64(0)
```

```python
users.duplicated().sum()
```

```
np.int64(0)
```

```python
ratings.duplicated().sum()
```

```
np.int64(0)
```

Popularity Based Recommender system Top 50 books

1. minimum 250 votes

```
1 ratings_with_name = ratings.merge(books, on='ISBN')
```

```
1 ratings_with_name.head(2)
```

	User-ID	ISBN	Book-Rating	Book-Title	Book-Author	Year-Of-Publication	Publisher	Image-URL-S	Image-URL-M	
0	276725	034545104X	0	Flesh Tones: A Novel	M. J. Rose	2002	Ballantine Books	http://images.amazon.com/images/P/034545104X.0...	http://images.amazon.com/images/P/034545104X.0...	http:/
1	276727	0446520802	0	The Notebook	Nicholas Sparks	1996	Warner Books	http://images.amazon.com/images/P/0446520802.0...	http://images.amazon.com/images/P/0446520802.0...	http:/

```
1 print("The shape of ratings with name : ",ratings_with_name.shape)
```

```
The shape of ratings with name :  (761568, 10)
```

```
1 ratings_with_name.groupby('Book-Title').count()
```

Show hidden output

```
1 num_rating_df = ratings_with_name.groupby('Book-Title').count()['Book-Rating'].reset_index()
2 num_rating_df.rename(columns={'Book-Rating':'num_ratings'},inplace=True)
3 num_rating_df
```

	Book-Title	num_ratings
0	A Light in the Storm: The Civil War Diary of ...	4
1	Beyond IBM: Leadership Marketing and Finance ...	1
2	Earth Prayers From around the World: 365 Pray...	10
3	Final Fantasy Anthology: Official Strategy Gu...	4
4	Good Wives: Image and Reality in the Lives of...	10
...	...	...
92513	Ã?Â?berallnie. AusgewÃ?Â¤hlte Gedichte 1928 - ...	1
92514	Ã?Â?bermorgen.	1
92515	Ã?Â?rger mit Produkt X. Roman.	4
92516	Ã?Â?stlich der Berge.	3
92517	Ã?Â?thique en toc	2

92518 rows × 2 columns

Average rating

```
ratings_with_name['Book-Rating'] = pd.to_numeric(ratings_with_name['Book-Rating'], errors='coerce')
avg_rating_df = ratings_with_name.groupby('Book-Title')['Book-Rating'].mean().reset_index()
avg_rating_df.rename(columns={'Book-Rating':'avg_rating'},inplace=True)
avg_rating_df
```

	Book-Title	avg_rating
0	A Light in the Storm: The Civil War Diary of ...	2.250000
1	Beyond IBM: Leadership Marketing and Finance ...	0.000000
2	Earth Prayers From around the World: 365 Pray...	5.000000
3	Final Fantasy Anthology: Official Strategy Gu...	5.000000
4	Good Wives: Image and Reality in the Lives of...	3.200000
...	...	...
92513	Ã?Â?berallnie. AusgewÃ?Â¤hlte Gedichte 1928 - ...	10.000000
92514	Ã?Â?bermorgen.	0.000000
92515	Ã?Â?rger mit Produkt X. Roman.	5.250000
92516	Ã?Â?stlich der Berge.	2.666667
92517	Ã?Â?thique en toc	4.000000

92518 rows × 2 columns

```python
popular_df = num_rating_df.merge(avg_rating_df,on='Book-Title')
popular_df

```

	Book-Title	num_ratings	avg_rating
0	A Light in the Storm: The Civil War Diary of ...	4	2.250000
1	Beyond IBM: Leadership Marketing and Finance ...	1	0.000000
2	Earth Prayers From around the World: 365 Pray...	10	5.000000
3	Final Fantasy Anthology: Official Strategy Gu...	4	5.000000
4	Good Wives: Image and Reality in the Lives of...	10	3.200000
...	...	...	...
92513	Ã?Â?berallnie. AusgewÃ?Â¤hlte Gedichte 1928 - ...	1	10.000000
92514	Ã?Â?bermorgen.	1	0.000000
92515	Ã?Â?rger mit Produkt X. Roman.	4	5.250000
92516	Ã?Â?stlich der Berge.	3	2.666667
92517	Ã?Â?thique en toc	2	4.000000

92518 rows × 3 columns

num rating > 250

```python
popular_df = popular_df[popular_df['num_ratings']>=250].sort_values('avg_rating',ascending=False).head(50)
```

```python
popular_df
```

Show hidden output

```python
popular_df = popular_df.merge(books,on='Book-Title').drop_duplicates('Book-Title')[
    ['Book-Title','Book-Author','Image-URL-M','num_ratings','avg_rating']
    ]
```

```python
popular_df
```

Show hidden output

1. user should be rating minimum = 200
2. books rating minimum = 50

collaboration filtering based recommender system

```python
ratings_with_name
```

Show hidden output

```python
x = ratings_with_name.groupby('User-ID').count()['Book-Rating'] > 200
experienced_users = x[x].index
```

```python
filtered_rating = ratings_with_name[ratings_with_name['User-ID'].isin(experienced_users)]
```

```python
filtered_rating
```

Show hidden output

```python
y = filtered_rating.groupby('Book-Title').count()['Book-Rating']>=50
famous_books = y[y].index
```

```
1 famous_books
```

```
Index(['1984', '1st to Die: A Novel', '2nd Chance', '4 Blondes',
       'A Bend in the Road', 'A Case of Need',
       'A Child Called \It\": One Child's Courage to Survive"',
       'A Civil Action', 'A Fine Balance',
       'A Heartbreaking Work of Staggering Genius',
       ...
       'Wild Animus', 'Winter Moon', 'Winter Solstice', 'Wish You Well',
       'Without Remorse', 'Wuthering Heights', 'You Belong To Me',
       'Zen and the Art of Motorcycle Maintenance: An Inquiry into Values',
       'Zoya', '\O\" Is for Outlaw"'],
      dtype='object', name='Book-Title', length=544)
```

```
1 final_ratings = filtered_rating[filtered_rating['Book-Title'].isin(famous_books)]
```

```
1 final_ratings
```

Show hidden output

```
1 print("The shape of final ratings : ",final_ratings.shape)
```

```
The shape of final ratings :  (42934, 10)
```

```
1  final_ratings.drop_duplicates()
```

	User-ID	ISBN	Book-Rating	Book-Title	Book-Author	Year-Of-Publication	Publisher	Image-URL-S	Image-
783	277427	002542730X	10	Politically Correct Bedtime Stories: Modern Ta...	James Finn Garner	1994	John Wiley & Sons Inc	http://images.amazon.com/images/P/002542730X.0...	http://images.amazon.com/images/P/00254273(
792	277427	0060930535	0	The Poisonwood Bible: A Novel	Barbara Kingsolver	1999	Perennial	http://images.amazon.com/images/P/0060930535.0...	http://images.amazon.com/images/P/00609305
794	277427	0060934417	0	Bel Canto: A Novel	Ann Patchett	2002	Perennial	http://images.amazon.com/images/P/0060934417.0...	http://images.amazon.com/images/P/00609344
796	277427	0061009059	9	One for the Money (Stephanie Plum Novels (Pape...	Janet Evanovich	1995	HarperTorch	http://images.amazon.com/images/P/0061009059.0...	http://images.amazon.com/images/P/00610090
801	277427	006440188X	0	The Secret Garden	Frances Hodgson Burnett	1998	HarperTrophy	http://images.amazon.com/images/P/006440188X.0...	http://images.amazon.com/images/P/00644018
...	...	...	...	...	...	...	...	...	
759529	275970	0804111359	0	Secret History	DONNA TARTT	1993	Ballantine Books	http://images.amazon.com/images/P/0804111359.0...	http://images.amazon.com/images/P/08041113
759530	275970	0804117683	0	Moo	Jane Smiley	1998	Ivy Books	http://images.amazon.com/images/P/0804117683.0...	http://images.amazon.com/images/P/08041176
759656	275970	140003065X	0	A Fine Balance	Rohinton Mistry	2001	Vintage Books USA	http://images.amazon.com/images/P/140003065X.0...	http://images.amazon.com/images/P/14000306
759660	275970	1400031354	0	Tears of the Giraffe (No.1 Ladies Detective Ag...	Alexander McCall Smith	2002	Anchor	http://images.amazon.com/images/P/1400031354.0...	http://images.amazon.com/images/P/14000313
759769	275970	1586210661	9	Me Talk Pretty One Day	David Sedaris	2001	Time Warner Audio Major	http://images.amazon.com/images/P/1586210661.0...	http://images.amazon.com/images/P/15862106

42934 rows × 10 columns

pivot table

```
pt = final_ratings.pivot_table(index='Book-Title',columns='User-ID',values='Book-Rating')
```

```
pt
```

User-ID	254	2276	2766	2977	3363	4017	4385	6251	6323	6543	...	271448	271705	273979	274004	274061	274301	274308	275970	277427	278418
Book-Title																					
1984	9.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	10.0	NaN	NaN	NaN	NaN	NaN	0.0	NaN	NaN
1st to Die: A Novel	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	9.0	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2nd Chance	NaN	10.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.0	...	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	NaN	NaN
4 Blondes	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	NaN	...	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
A Bend in the Road	0.0	NaN	7.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Wuthering Heights	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	NaN	NaN
You Belong To Me	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	...	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Zen and the Art of Motorcycle Maintenance: An Inquiry into Values	NaN	NaN	NaN	NaN	0.0	NaN	NaN	0.0	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	NaN
Zoya	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
\O\" Is for Outlaw"	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	8.0	NaN	NaN	NaN	NaN

544 rows × 569 columns

```
pt.fillna(0,inplace=True)
```

```
pt
```

User-ID Book-Title	254	2276	2766	2977	3363	4017	4385	6251	6323	6543	...	271448	271705	273979	274004	274061	274301	274308	275970	277427	278418
1984	9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st to Die: A Novel	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd Chance	0.0	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4 Blondes	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A Bend in the Road	0.0	0.0	7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Wuthering Heights	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
You Belong To Me	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Zen and the Art of Motorcycle Maintenance: An Inquiry into Values	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Zoya	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
\O\" Is for Outlaw"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	8.0	0.0	0.0	0.0	0.0

544 rows × 569 columns

cosine similarity

```
from sklearn.metrics.pairwise import cosine_similarity
```

```
similarity_scores = cosine_similarity(pt)
```

```
similarity_scores.shape
```

```
(544, 544)
```

```
similarity_scores[0]
```

Show hidden output

```
list(enumerate(similarity_scores[0]))
# distance + items
```

Show hidden output

```
# sorted based on similarity score
sorted(list(enumerate(similarity_scores[0])),key=lambda x:x[1],reverse=True)
```

Show hidden output

```
# similar book top 1 to 6
sorted(list(enumerate(similarity_scores[0])),key=lambda x:x[1],reverse=True)[1:6]
```

```
[(409, np.float64(0.33461279154172696)),
 (364, np.float64(0.2703889957883732)),
 (394, np.float64(0.2603543880046459)),
 (129, np.float64(0.25202417820000605)),
 (374, np.float64(0.2431472885801574))]
```

```
pt.index.get_loc('Zoya')
```

```
542
```

create function for recommend

```
np.where(pt.index=='Zoya')[0][0]
```

```
np.int64(542)
```

```
np.where(pt.index=='1984')[0][0] # find the book index
```

```
np.int64(0)
```

```
# create function for recommendation

def recommend(book_name):
  index = np.where(pt.index==book_name)[0][0]
  similar_items = sorted(list(enumerate(similarity_scores[index])),key=lambda x:x[1],reverse=True)[1:6]

  for i in similar_items:
    print(pt.index[i[0]])
```

```
recommend('Wuthering Heights')
```

```
Pleading Guilty
The Thorn Birds
Primary Colors: A Novel of Politics
Dances With Wolves
Like Water for Chocolate: A Novel in Monthly Installments With Recipes, Romances and Home Remedies
```

```
recommend('Zoya')
```

```
Secrets
Kaleidoscope
Heartbeat
GARDEN OF SHADOWS (Dollanger Saga (Paperback))
Wanderlust
```

```
recommend('1984')
```

```
The Handmaid's Tale
The Bonesetter's Daughter
The Fellowship of the Ring (The Lord of the Rings, Part 1)
Fahrenheit 451
The Catcher in the Rye
```

image url

```
popular_df['Image-URL-M'][0]
```

```
'http://images.amazon.com/images/P/0439136350.01.MZZZZZZZ.jpg'
```

pickle

```
import pickle
pickle.dump(popular_df,open('popular.pkl','wb'))

```

```
import streamlit as st
import pickle
import pandas as pd

# Page configuration
st.set_page_config(page_title="Book Recommender", layout="wide")

```

```
# Load data
@st.cache_data
def load_data():
    return pickle.load(open('popular.pkl', 'rb'))

popular_df = load_data()

# --- Advanced CSS for UI Customization ---
st.markdown("""
    <style>
    /* 1. BOLD TABS */
    button[data-baseweb="tab"] p {
        font-size: 18px !important;
        font-weight: bold !important;
        color: #1e293b !important;
    }

    /* 2. CENTERED RECOMMEND BUTTON */
    .stButton {
        display: flex;
        justify-content: center;
        margin-top: 20px;
    }

    /* 3. RECOMMEND BUTTON CSS */
    .stButton > button {
        background-color: #4F46E5 !important;
        color: white !important;
        border-radius: 20px !important;
        padding: 10px 30px !important;
        border: none !important;
        font-weight: bold !important;
        transition: 0.3s;
    }

    /* 4. SMALLER OUTPUT BOXES */
    .rec-card {
        background-color: white;
        padding: 10px;
```

```
        border-radius: 8px;
        box-shadow: 0 2px 8px rgba(0,0,0,0.1);
        text-align: center;
        width: 100%;
        margin: auto;
    }
    .rec-img {
        width: 100%;
        height: 160px;
        object-fit: contain;
        border-radius: 4px;
    }

    /* 5. CUSTOM FOOTER CSS */
    .footer {
        position: relative;
        left: 0;
        bottom: 0;
        width: 100%;
        background-color: transparent;
        color: #64748b;
        text-align: center;
        padding: 20px;
        font-family: 'Trebuchet MS', sans-serif;
        font-size: 14px;
        font-weight: bold;
        border-top: 1px solid #e2e8f0;
        margin-top: 50px;
        letter-spacing: 1px;
    }
    .footer span {
        color: #ef4444; /* Red heart color */
    }

    /* Home Card Styling */
    .book-card {
        background-color: white;
        padding: 15px;
        border-radius: 12px;
```

```
        box-shadow: 0 4px 15px rgba(0,0,0,0.1);
        text-align: center;
        margin-bottom: 20px;
    }
    </style>
    """, unsafe_allow_html=True)

st.markdown("<h1 style='text-align: center;'>📖 Book Hub</h1>", unsafe_allow_html=True)
tab1, tab2 = st.tabs(["🏠 Home", "🔍 Recommend"])

# --- TAB 1: HOME ---
with tab1:
    st.markdown("### Top Trending Books")
    cols_per_row = 5
    for i in range(0, 50, cols_per_row):
        cols = st.columns(cols_per_row)
        batch = popular_df.iloc[i : i + cols_per_row]
        for col, (index, row) in zip(cols, batch.iterrows()):
            with col:
                st.markdown(f"""
                    <div class="book-card">
                        <img src="{row['Image-URL-M']}" style="width:100%; border-radius:8px;">
                        <div style="font-weight:bold; margin-top:10px; font-size:14px;">{row['Book-Title']}</di
                        <div style="color:gray; font-size:11px;">{row['Book-Author']}</div>
                    </div>
                    """, unsafe_allow_html=True)

# --- TAB 2: RECOMMEND ---
with tab2:
    st.markdown("<h3 style='text-align: center;'>Find Your Next Read</h3>", unsafe_allow_html=True)
    book_list = popular_df['Book-Title'].unique()
    selected_book = st.selectbox("Search Book", book_list, index=None, label_visibility="collapsed")

    if st.button("Get Recommendations"):
        if selected_book:
            st.markdown(f"<p style='text-align:center;'>Top picks for fans of <b>{selected_book}</b></p>", unsa
            res_cols = st.columns(6)
            sample_results = popular_df.sample(6)
            for col, (idx, row) in zip(res_cols, sample_results.iterrows()):
```

```
                with col:
                    st.markdown(f"""
                        <div class="rec-card">
                            <img src="{row['Image-URL-M']}" class="rec-img">
                            <div style="font-size:11px; font-weight:bold; margin-top:5px; height:30px; overflow
                        </div>
                        """, unsafe_allow_html=True)
        else:
            st.warning("Please select a book first!")

# --- BEAUTIFUL CUSTOM FOOTER ---
st.markdown("""
    <div class="footer">
        Designed & Developed with <span>❤</span> by Yanaguntikar Meesal
    </div>
    """, unsafe_allow_html=True)
```